Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the energy system and assumptions.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your model or result. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

A 3 kg object descends 5 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$. Mechanical energy conserved when only conservative interactions transfer energy. Use g = 10 N/kg near the surface.

Worked reasoning: A 1 kg object drops 5 m from rest with g=10 m/s² and no energy loss. It loses 50 J of potential energy and gains 50 J of kinetic energy, giving speed 10 m/s.

Your turn: Draw energy bars at the start and just before arrival.
Label the system and neglected losses.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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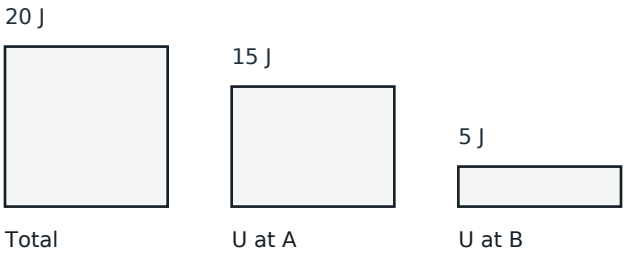
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A system has total mechanical energy 20 J. At one point U=15 J; at another U=5 J.
Find K at each point.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg ball starts at height 5 m with speed 4 m/s.
Find its speed at height 1 m.
Use g = 10 N/kg and ignore air resistance.
Label the height reference and complete the energy account first.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail?
Name the destination to include.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A and B have equal momentum magnitude, but A has 4 times B’s mass.
Compare speeds and kinetic energies algebraically.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, including direction

Sentence starter: Both … . They differ in … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A ball is thrown upward at 10 m/s from h=0.
Find maximum height without drag, using g=10.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2 kg cart starts at 4 m/s and rises 3 m without friction; g=10. Can it reach that height?
Find its maximum rise.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3